Econometric Game 2026
Case B

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Center for Research in Energy
ECONOMICS AND MARKETS

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1 Instructions for structuring the report

Your report should be clear, structured, and well-argued, demonstrating both methodological rigor and economic intuition. The volume of the report should not exceed 25 pages including Appendix. Below is the recommended structure:

1. Introduction

   Briefly introduce the problem and its economic relevance. State your research questions and objectives. Outline the structure of the report.

2. Data Description

   Provide relevant descriptive statistics for selected variables in the dataset that you consider particularly relevant for the analysis. What are the main features of the different time series, such as volatility and price span, periodicity, seasonality, etc.? Explain key variables, transformations, and any data cleaning steps. Highlight with appropriate graphs and tables.

3. Methodology

   Clearly explain the econometric model(s) applied. Justify the choice of estimation and forecasting techniques. Discuss any assumptions and potential econometric issues.

4. Empirical Results

   Present key estimation and forecasting results in tables and figures and conduct appropriate interpretations. Perform robustness checks and discuss potential limitations.

5. Discussion and Economic Interpretation

   Relate and interpret your findings for real-world implications and refer where relevant to economic theory including implications for electricity trading over various horizons. Discuss any policy recommendations or business implications if relevant.

6. Conclusion

   Summarize key findings concisely. Highlight limitations and potential extensions for future research.

7. References

   List all academic papers, datasets, and other sources cited in a consistent format. If using ChatGPT or similar as a reference source, make the appropriate reference and possibly with quotation marks.

8. Statement about the use of hardware and software

   There are no restrictions on the use of hardware or software. However, use of cloud computing resources are not permitted. The report should include a statement about **all** the computing software that has been used to work with the Case. Note particular guidelines about using ChatGPT and other AI tools.

9. Statement about use of ChatGPT

   Compulsory: In your report, your reference list should include a statement explaining exactly how ChatGPT and other AI tools have been used in working with your report. If the statement is absent, the report will be withdrawn from assessment. See also section 4 below.

10. Appendix (if necessary)

    Include additional tables, robustness tests, or technical derivations that support the main text.

2 Instructions for structuring the presentation

1. Introduction

   Brief introduction (1-2 lines) to the research topic. Goal: Set the context and the research question for your audience.

2. Methodology

   Outline the methodology used in the analysis or research. Keep it simple but informative.

3. Summary statistics

   Present key summary statistics. Brief commentary on what these statistics reveal about the data.

4. Main results

   Present the main findings or results of your analysis. Be concise and focus on the most important outcomes.

5. Interpretation, policy recommendations, Conclusion

   Provide a brief interpretation of the results. What do the findings mean in a real-world context?

   Offer 2-3 policy recommendations or actionable insights based on the results and conclude.

Tips:

- **Stick to a 1-minute per slide rule. You have a 5-minute limit for the presentation.**
- Keep text minimal and focused.
- Use visuals to highlight key points and keep the audience engaged.
- Use Google Slides to prepare your slides.

3 Instructions for submitting the report and slides

- The deadline for the final report and presentation slides submission is today at 17:00. You should do so by sending the files to juryeg@vsae.nl. Subject of this email: "Final Case B - [University Name]
- The total volume of the report including Appendix should not exceed 25 pages. Be mindful of page formatting: use standard font size (e.g., 12 pt), appropriate margins (e.g., 1 inch), and spacing (e.g., 1.5 or 2x line spacing).

4 Guidelines for using ChatGPT and other AI tools

- Allowed Usage:
  ChatGPT can be used for instance to assist with coding tasks (including debugging), optimization, and understanding programming concepts. You may use ChatGPT to generate code snippets, explore different coding approaches, and troubleshoot errors.

- Not Allowed Usage:
  ChatGPT cannot be used to *write* your report, including explanations, analysis, or conclusions. All written content in your report must be your own work, based on your understanding and research. If you use ChatGPT as a source for your report, you should indicate this explicitly by appropriate referencing like any other references.

- **Compulsory: In your report, your reference list should include a statement explaining exactly how ChatGPT and other AI tools have been used in working with your report. If the statement is absent, the report will be withdrawn from assessment.**

5 Case B

The electricity market is changing rapidly. The green transition is reshaping the role of renewable sources in electricity production, and this shift is substantially transforming the market. In particular, the rise of renewable energy has influenced both electricity prices and their volatility. Changes to the transmission grid also strongly affect how prices evolve, as markets become more interconnected and efficient.[1] The grid allows producers to move electricity across Europe—from areas with high production to those with lower production. This has become more important than ever, since renewable energy sources often generate more power than a region needs at a given moment. The combination of more renewable generation and a more interconnected grid means that the work of an electricity price forecaster today is quite different from what it was ten years ago, as price forecasting is now inherently a multivariate problem.

You will be provided with a dataset containing the same variables as before, but also for the other price zone of Denmark (DK2), Norway (NO2), Sweden (SE4) and Germany (DE). You will also be provided with the transmission data on the flows between these price zones. Finally, you will also be provided with daily weather data collected in several meteorological stations in Europe. In particular, such dataset includes precipitation amount, cloud cover, sunshine duration, mean temperature and wind speed. With such a dataset your task is to address the following points,

- Produce one step ahead forecasts for all hourly spot prices for price zone DK1 (the same as for case A) for all days in 2024. *Note: This is substantially different from Case A.* The forecast for all 24 hourly prices for the next day will be produced using information available up to the current day, over the whole year span. This amounts to (365×24) forecasts for each price zone. Repeat the same exercise for the year 2020.

- Highlight the change, if there was any, in the role of renewable sources production in terms of predictive power over the considered horizon. Is there any difference in the production over such a time span? Is there any difference in the impact that these production sources have on the price dynamics?

- Highlight the presence, if any, of network effects, in the considered electricity market. Is there any spillover across price zones? Do variables across price zones have predictive or explanatory power for electricity prices? Did the network structure of electricity around Denmark change substantially in its dynamics and impact on electricity price over the considered time span? Present your findings with appropriate graphs and tables.

Remark. *Note that the dataset provided is substantially larger than what is provided in Case A. You are required to consider several price zones. However,*

[1] Have a look at the Nordpool Group and Energinet websites for more insights into the production and transmission networks for the relevant price zones.

you are not required to use all the provided data. It is up to you to select what is more helpful to produce the best forecast.

6 Additional Data Description

Climate data

This folder contains daily weather observations from the *European Climate Assessment & Dataset (ECA&D)*. Observations are provided at the station level and then *assigned* to electricity bidding zones (e.g., DK1, DK2, NO2, DE, SE4) using station coordinates. Each file contains one row per station-day (i.e., at most one observation per station and calendar day in the final datasets). Only observations flagged as *valid* by ECA&D quality control are retained; consequently, some stations have missing days.[2]

For each bidding zone there is a separate compressed CSV file (`.csv.gz`) for each climate variable: `cc` (cloud cover), `tg` (mean temperature), `rr` (precipitation), `ss` (sunshine duration), and `fg` (mean wind speed). Note that DK1 and DK2 do not contain cloud cover (`cc`) in the provided sample.

Columns (shared across all climate files). All climate files share the following columns:

- `DATE`: Observation date (daily frequency; formatted as `YYYY-MM-DD` in the exported files).
- `staid`: ECA&D station identifier.
- `station_name`: Station name.
- `ZONE_ID`: Electricity bidding zone to which the station is assigned.
- `LAT_DEC`, `LON_DEC`: Station latitude/longitude in decimal degrees. ECA&D stores coordinates in degrees–minutes–seconds (DMS), e.g. `+55:51:00`, which we convert to decimal degrees using

$$\text{decimal} = \text{sign} \times \left(\text{deg} + \frac{\text{min}}{60} + \frac{\text{sec}}{3600} \right).$$

- `elevation_m`: Station elevation in meters.
- `ELEID`: ECA&D element identifier describing the measurement/aggregation convention used for the daily value (e.g., the time window and averaging/accumulation rule). The set of possible `ELEID`s differs by variable and may vary across stations.

[2] More information on ECA&D is available at `https://www.ecad.eu`.

Variable-specific columns and element identifiers. In addition to the shared columns above, each file contains a single variable column (last column) with the daily value. The `ELEID` indicates the ECA&D aggregation/measurement convention used for that station-day; for a given variable, multiple `ELEID`s may occur across stations and zones.

Cloud cover (`cc`)

- **Element identifiers (ELEID) observed in the datasets:**
  - **CC1:** Mean daily cloud cover, 0–0 UT (unit: oktas)
  - **CC3:** Mean daily cloud cover, unknown interval (unit: oktas)
  - **CC6:** Mean daily cloud cover, mean of 7, 14 and 21 UTC (unit: oktas)
  - **CC7:** Mean daily cloud cover, 18–18 UT (unit: oktas)
  - **CC13:** Mean daily cloud cover, average of 6, 12 and 18 UTC (unit: oktas)
  - **CC23:** Mean daily cloud cover, average of 7, 14 and 19 CET (unit: oktas)
- **Last column `cc`:** Mean daily cloud cover (oktas).

Mean temperature (`tg`)

- **Element identifiers (ELEID) observed in the datasets:**
  - **TG3:** Mean temperature 0–0 UT (unit: 0.1 °C)
  - **TG5:** Mean temperature calculated as average of TN and TX (unit: 0.1 °C)
  - **TG6:** Mean temperature calculated as weighted average of TN, TX and observations at 06, 12 and 18 UT (5 values) (unit: 0.1 °C)
  - **TG7:** Mean temperature calculated as average of 3 or more observations each day (unit: 0.1 °C)
  - **TG11:** Mean temperature 18–18 UT (unit: 0.1 °C)
  - **TG14:** Mean temperature calculated as average of 07, 14, 21, 21 CET (unit: 0.1 °C)
  - **TG21:** Mean temperature calculated as average of 24 hourly values (0–0 UTC) (unit: 0.1 °C)
  - **TG23:** Mean temperature 0–0 local time (with changes to winter and summer time) (unit: 0.1 °C)
- **Last column `tg`:** Mean daily temperature (0.1 °C).

Precipitation (`rr`)

- **Element identifiers (ELEID) observed in the datasets:**

  - **RR2:** Precipitation amount morning previous day 06,07,08,09 until morning today (shifted 1 day back by ECA staff) (unit: 0.1 mm)
  - **RR3:** Precipitation amount morning today 06,07,08 until morning next day (unit: 0.1 mm)
  - **RR7:** Precipitation amount 0–0 UT (unit: 0.1 mm)
  - **RR8:** Sum of 12-hourly precipitation of observations at 18 UT today and 6 UT tomorrow (2 values) (unit: 0.1 mm)
  - **RR9:** Precipitation amount morning today 06:00 UTC until morning next day (unit: 0.1 mm)
  - **RR13:** Precipitation amount 22–22 or 23–23 UT (unit: 0.1 mm)
  - **RR18:** Precipitation amount morning today 07:00 CET until morning next day (unit: 0.1 mm)
  - **RR19:** Precipitation amount 0–0 local time (with changes to winter and summer time) (unit: 0.1 mm)

- **Last column `rr`:** Daily precipitation amount (0.1 mm).

Sunshine duration (`ss`)

- **Element identifiers (ELEID) observed in the datasets:**
  - **SS3:** Daily sunshine, 0–0 UT (unit: 0.1 Hours)
  - **SS4:** Daily sunshine 18–18 UT (unit: 0.1 Hours)
  - **SS6:** Daily sunshine 0–0 local time (with changes to winter and summer time) (unit: 0.1 Hours)

- **Last column `ss`:** Daily sunshine duration (0.1 hours).

Mean wind speed (`fg`)

- **Element identifiers (ELEID) observed in the datasets:**
  - **FG1:** Average wind speed of 24 hourly measurements of 10-min averages (0–0 UT) (unit: 0.1 m/s)
  - **FG2:** Average wind speed of measurements at 06, 12, 18, (00) UTC (unit: 0.1 m/s)
  - **FG4:** Average wind speed, mean of 7, 14 and 21 CET (unit: 0.1 m/s)
  - **FG5:** Average wind speed 18–18 UT, 8 three-hourly observations (unit: 0.1 m/s)
  - **FG7:** Average wind speed of 24 hourly measurements (6 ten-minute averages per hour), 0–0 UT (unit: 0.1 m/s)
  - **FG17:** Average wind speed, unknown interval (unit: 0.1 m/s)

  - **FG18:** Average wind speed, mean of 7, 14 and 19 CET (unit: 0.1 m/s)
  - **FG19:** Average wind speed 0–0 local time (with changes to winter and summer time) (unit: 0.1 m/s)

- **Last column `fg`:** Mean wind speed (0.1 m/s).